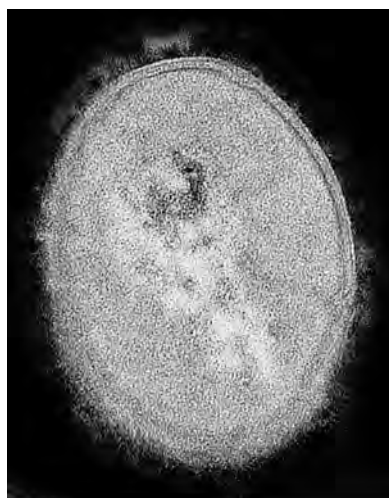


0 2

Figure 1 shows a transmission electron microscope (TEM) image of a *Streptococcus* bacterium.

Figure 1



0 2 . 1

Describe how the appearance of the area containing DNA in a TEM image of a eukaryotic cell would differ from that shown in **Figure 1**.

[2 marks]

0 2 . 2

Describe **one** difference between the structure of DNA in a prokaryotic cell and in a eukaryotic cell.

[1 mark]

Question 2 continues on the next page

Turn over ►



0 2 . 3

Streptococcus bacteria can infect the lungs when air is breathed in and cause lung disease.

Describe the mechanism of breathing that causes air to enter the lungs.

[3 marks]

0 2 . 4

Some strains of *Streptococcus* bacteria are more likely to cause lung disease than other strains.

Strains that do not cause lung disease are quickly destroyed by phagocytes. Phagocytes are stimulated when they bind to murein on *Streptococcus* bacteria.

Each strain of *Streptococcus* bacteria has a capsule of different thickness from the others.

Suggest how *Streptococcus* bacteria with a thicker capsule are more likely to survive **and** so cause lung disease.

[2 marks]

